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Perspective

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Accelerated Approval and Expensive Drugs — A Challenging Combination

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For serious or life-threatening disease, the Food and Drug Administration (FDA) can approve drugs on the basis of surrogate end points that are "reasonably likely to predict clinical benefit," through its accelerated approval review track. This pathway, which dates back to the early 1990s, was designed as a response to the demand for faster drug development in the context of the HIV/AIDS crisis.[1] Since then, the accelerated-approval program has expanded to include oncology products and drugs for other diseases, now accounting for about 10% of new drug approvals.[2]

The accelerated pathway commits a drug manufacturer to conducting postapproval trials to confirm the drug's efficacy for the approved indication. Manufacturers must also include in the official labeling a statement that the clinical benefit "has not been established" and that confirmatory trials are ongoing; the labeling is revised when those trials are complete.

There has been positive early experience with accelerated approval,[1] with some notable exceptions. For example, the FDA authorized marketing of bevacizumab (Avastin) for breast cancer on the basis of the surrogate measure of progression-free survival, but then follow-up trials showed no increase in overall survival and raised important safety concerns. The indication was eventually removed after a year of hearings and debate.

Despite the uncertain evidence of clinical benefit, drugs receiving accelerated approval enter the market as FDA-approved products, and insurers must decide whether and how to pay for them. Those decisions are becoming increasingly complex in light of the rising prices of new drugs (see table). For example, eteplirsen (Exondys 51) was given accelerated approval for use in muscular dystrophy in September 2016 on the basis of very small increases in the production of dystrophin protein; controversy ensued about whether such increases could actually affect clinical progression of the disease. When it received the approval, the manufacturer announced a price of $300,000 per year, or more depending on the patient's weight. Initial optimism about expanding access to this new therapy was suddenly met with the stark reali-

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Selected Drugs That Have Received Accelerated Approval since 2011 and Their Listed Cost.*

Approval Year	Drug (Brand Name)	Initial Indication	Surrogate Measure Used for Approval	Current Cost ($/mo)
2011	Crizotinib (Xalkori)	Anaplastic lymphoma kinase (ALK)–positive locally advanced or metastatic non–small-cell lung cancer	Overall response rate based on Response Evaluation Criteria in Solid Tumors	14,353
2012	Bedaquiline (Sirturo)	In combination therapy for pulmonary multidrug-resistant tuberculosis	Time to sputum culture conversion	6,000
2013	Pomalidomide (Pomalyst)	Multiple myeloma that has progressed despite receipt of two prior therapies	Overall response rate, based on European Group for Blood and Marrow Transplant criteria	14,165
2014	Blinatumomab (Blincyto)	Philadelphia chromosome–negative relapsed or refractory B-cell acute lymphoblastic leukemia	Complete remission or complete remission with partial hematologic recovery rate	56,262
2014	Pembrolizumab (Keytruda)	Unresectable or metastatic melanoma with disease progression	Overall response rate based on Response Evaluation Criteria in Solid Tumors	9,252
2014	Ceritinib (Zykadia)	ALK-positive locally advanced or metastatic non–small-cell lung cancer with disease progression or intolerance to crizotinib	Overall response rate based on Response Evaluation Criteria in Solid Tumors	14,628
2015	Panobinostat (Farydak)	Multiple myeloma that has progressed despite receipt of two prior therapies	Progression-free survival based on European Group for Blood and Marrow Transplant criteria	10,625
2015	Palbociclib (Ibrance)	Postmenopausal women with metastatic estrogen receptor–positive, human epidermal growth factor receptor 2–negative advanced breast cancer	Progression-free survival based on Response Evaluation Criteria in Solid Tumors	11,224
2016	Eteplirsen (Exondys 51)	Duchenne's muscular dystrophy in patients with confirmed mutation amenable to exon 51 skipping	Increase in dystrophin in skeletal muscle	57,600

* Initial indications and surrogate measures were obtained from drug labeling (www.accessdata.fda.gov/scripts/cder/daf/index.cfm). Costs of drugs were obtained from DrugAbacus (run by the Memorial Sloan Kettering Cancer Center) in November 2016, except for the cost of bedaquiline, which is based on the average wholesale price (AWP) of $36,000 for 24 weeks of treatment (as reported in UpToDate), and the cost of eteplirsen, which is based on the AWP of $1,920 per 100 mg (assuming a patient weighing 25 kg and a dose of 30 mg per kilogram of body weight once weekly), or $691,000 per year. Publicly reported estimates of $300,000 included presumed discounts.

ties of paying for it. One national insurer, Anthem, declined to cover the drug at the time of approval, calling it an "experimental" treatment. Another, Humana, decided to cover it only for patients who were still ambulatory and imposed criteria for discontinuation based on clinical response.

In contrast to private insurers, Medicaid and Medicare programs operate under more stringent rules about what therapies they must cover. Medicaid has a statutory requirement to cover nearly all FDA-approved drugs, with hardly any cost sharing by patients. Medicaid programs can institute prior-authorization requirements to ensure that a prescription is medically indicated, but limiting access to approved drugs in high demand is difficult. When some Medicaid programs tried to limit coverage of the expensive direct-acting antiviral drugs for hepatitis C to patients who were at highest risk for harm, they were successfully sued for imposing allegedly improper prior-authorization requirements.

Medicare is required to cover drugs that are "reasonable and necessary," and private Part D plans have some flexibility in what they include on their formularies. The notable exceptions are drugs in six "protected classes," for which plans must include all or nearly all FDA-approved drugs. Cancer drugs are a protected class, and it is also the class that most commonly qualifies for accelerated approval.

Thus, at a time when medications can cost hundreds of thousands of dollars per year, accelerated approval can lead to

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situations in which private payers may choose not to cover a drug because of high cost and lack of evidence of clinical efficacy, thereby thwarting the pathway's goal of getting potentially important therapies to patients earlier, while major government payers are forced to cover the product, directing substantial tax dollars to drugs not yet shown to have clinical benefit. For public insurers, accelerated approval thus becomes not only a pathway for a new product to enter the market, but also a mandate to pay high prices for an unproven therapy.

We propose several policy options that could address this challenge. First, manufacturers could be required to offer additional price concessions to public insurance programs for drugs receiving accelerated approval until the confirmatory trials are completed. We see the value in getting products that qualify for accelerated approval to market for patients, but we believe the price paid by taxpayers should reflect the strength of the available evidence about the drug's clinical impact. For example, Medicaid is supposed to receive the "best price" the manufacturer offers to other payers, but an additional rebate could be statutorily added for accelerated-approval products; this would last until confirmation of the drug's clinical efficacy. Other options include holding a portion of the drug payment in escrow until the drug's efficacy is confirmed and then reimbursing payers if the drug is later found to be clinically ineffective.

Alternatively, to address the concern that additional required discounts will simply lead to higher list prices, companies could be reimbursed by government payers only for the cost of manufacturing the drug plus some agreed-on markup, until confirmatory trials demonstrate clinical benefit. We think these strategies would allow drugs to reach patients while also offering incentives to manufacturers to complete their postapproval commitments promptly.

Second, we would like to see the FDA do more to ensure that confirmatory trials conducted after receiving accelerated approval are performed in a timely fashion and are designed optimally to limit the period of uncertainty about true clinical effect. For example, when bedaquiline (Sirturo), an antibiotic for multidrug-resistant pulmonary tuberculosis, was approved on the accelerated track in 2012, the confirmatory trial was not set to be completed until 2022.[3] There are also examples among oncology products of confirmatory trials that have not been completed for 5 years or more after accelerated approval.[4] We believe there should be plans in place to begin confirmatory trials within 3 months after approval, with tracking of trial progress through ClinicalTrials.gov. Though the rarity of the disease and other factors might reasonably affect trial accrual times, there should also be meaningful repercussions for missing milestones such as having a protocol in place or hitting recruitment targets, culminating in withdrawal of the drug if the trial is unnecessarily delayed for an extended period. The FDA can, under current law, assess financial penalties or withdraw an accelerated-approval drug from the market if the manufacturer fails to conduct its confirmatory trial or fails to do so with "due diligence," a benchmark that the FDA can further clarify with stakeholder input.

For public insurers, accelerated approval thus becomes not only a pathway for a new product to enter the market, but also a mandate to pay high prices for an unproven therapy.

Finally, we propose that all drugs that move through the accelerated-approval pathway and cost over $100,000 per year, or some other agreed-on threshold, should be the subject of formal economic impact analyses after 1 to 2 years on the market. These could be performed by economists at the Centers for Medicare and Medicaid Services or the Government Accountability Office, possibly funded by an increase in the user fees for manufacturers that use this pathway. If a drug's high cost leads to financial turmoil in government insurance programs, then the value of accelerated approval is lost.

In an era of constrained resources at both the state and federal levels, accelerated approval of expensive drugs leads to reduced resources available for other health care services that are already supported by strong clinical evidence. We believe that changes to the

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accelerated-approval process are required in order to ensure that it continues to benefit society.

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Emerging from EHR Purgatory — Moving from Process to Outcomes

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The widespread adoption of electronic health records (EHRs) during the past decade has been hailed as a major advance in medical practice. Recently, however, a growing number of clinicians have spoken out about the counterproductive effects of these systems on patient care.[1,2] The national push toward greater implementation of EHRs was inspired by accumulating evidence that promised improvements in care coordination, quality, safety, and patient engagement. What has gone wrong?

The answer is, of course, multifaceted, but largely ignored in the discussion has been the effect of the underlying mode of physician payment. Payment at the clinician level remains predominantly fee for service, increasingly supplemented by pay for performance in an effort to increase value-based reimbursement. Both modes of payment are largely process-based and retrospective. EHRs are ideally suited for such process-based reimbursement since they provide detailed, piecemeal, readily audited documentation of process and volume. The more detailed the documentation of process, the greater the opportunity for reimbursement.

An electronic arms race has broken out as payers demand increasingly detailed documentation to justify payment and EHR vendors respond with ever more elaborate documentation tools — with payers and provider organizations spending billions of dollars in the process. Much to the chagrin of clinicians, the EHR has become a billing instrument that requires excessive process documentation, coding, and other activities that are devoid of meaning for patient care.

This state of affairs has become problematic for both doctors and patients. Extraneous EHR-focused demands consume precious visit time. Clinicians find themselves interacting more with their computers than with their patients, which compromises the patient care experience, productivity, and professional satisfaction.[1,2] Revenue per visit might be increasing, but at a cost of accommodating fewer visits and declining morale. Some observers have worried about an adverse effect on patient safety when doctors are distracted by the computer.[1] Primary care physicians find themselves especially overburdened because of the documentation demands for patients with multiple complex problems; staying late or taking work home has become the rule rather than the exception.[2]

A host of short-term solutions have been implemented to ease physicians' documentation-related workload, including voice-recognition software and the use of scribes and other nonphysician team members to enter notes. As attractive and necessary as these solutions may be in the short run, they are costly and imperfect; moreover, they don't address the root cause of excessive documentation requirements: the emphasis on process-based payment and the distorted use of the EHR as a billing instrument. It is ironic that process documentation and

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